INCH-POUND

MIL-G-12327J
6 June 1989
SUPERSEDING
MIL-G-0012327H(EA)
18 June 1984
MIL-G-12327E
18 September 1968

MILITARY SPECIFICATION

GRENADE, HAND, SMOKE, HC, M8

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hand grenade for use in producing white smoke.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1330

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SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood Preservative: Water Repellent
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-B-2427	-	Boxes, Ammunition Packing: Wood, Nailed
MIL-B-46506	-	Boxes, Ammunition Packing, Wood, Wirebound

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defectives
MIL-STD-1168	-	Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

DRAWINGS

19-48-4116/65	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets

(Copies are available from Director, U.S. Army Defense Ammunition Center and School, ATTN: SMCAC-DEO, Savanna, IL 61074-9639.)

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U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

13-9-105	-	Label, Continuous
13-9-120	-	Packing, Grenade
13-19-32	-	Grenade, Hand, Smoke, HC, AN-M8
13-19-38	-	Card, Instruction
13-19-83	-	Marking For: M8 and M18 Smoke Grenades
13-19-94	-	Starter Cup Assembly
143-1-1	-	White Smoke Mixture - I

(Copies are available from Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-O, Aberdeen Proving Ground, MD 21010-5423.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 13-19-32 and subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the grenade shall conform to the specifications and drawings listed on Drawing 13-19-32 and on subsidiary drawings.

3.2 Filling. The grenade shall be filled with HC smoke mixture as specified on Drawing 13-19-32.

3.3 Assembly. The grenade shall be assembled as shown on Drawing 13-19-32 (see 6.6).

3.4 Functioning. When tested in accordance with 4.2.4.1, the grenade shall ignite and emit a continuous smoke cloud under pressure within 15 seconds (lag time) after the fuze is activated. The smoke cloud shall be emitted under pressure (smoke emission time) for a period between 90 and 135 seconds. The total flaming time, if any, of the grenade shall

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not exceed 20 percent of the total burning time (see figure 1). The grenade shall show no evidence of deflagration [see 6.5(e)].

3.5 Volatiles content. The volatiles content of the starter mix slug shall not exceed 1 percent when tested in accordance with 4.2.4.2.

3.6 Moisture content. Each batch of pyromixture shall comply with Drawing 143-1-1 in accordance with 4.2.4.4.

3.7 Workmanship. The grenade shall be free from smoke mix contamination, foreign matter and damage such as dents, cracks or deformation which could impair its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

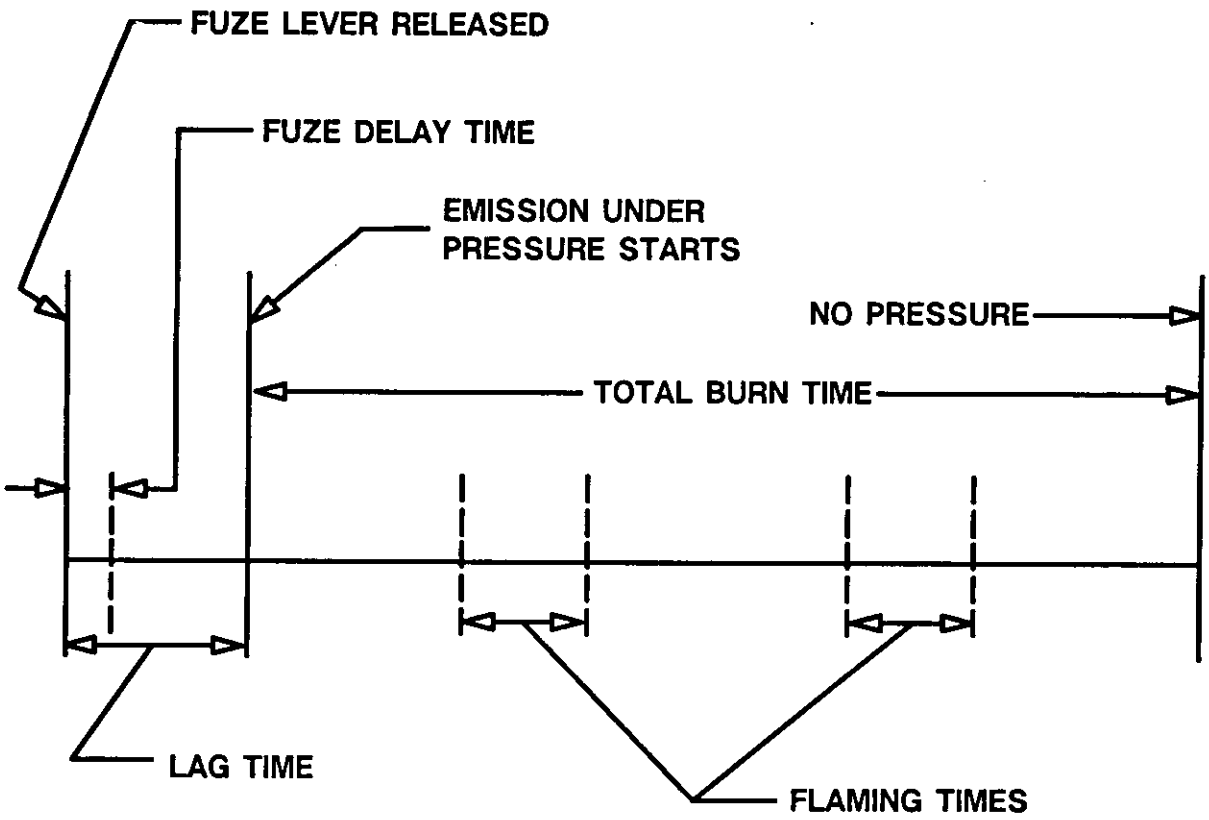
4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting.

4.2.1.1 Grenades. A lot shall be identified and controlled in accordance with MIL-STD-1168 and shall be produced by one manufacturer, in no more than five consecutive

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NOTE: SMOKE EMISSION TIME EQUALS TOTAL BURNING TIME MINUS TOTAL FLAMING TIME

FIGURE 1. Total burning time

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work shifts not to exceed two weeks consecutive duration, whichever occurs first, from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of fuzes or more than one interfix of metal components, smoke, starter mix ingredients, or starter slugs from one manufacturer shall be represented in any one lot of assembled grenades. The minimum lot size shall be 1,201 grenades.

4.2.1.2 Starter slugs. A lot of starter slugs shall consist of the starter slugs provided by one manufacturer using the same manufacturing process. Only one lot of each starter slug mix ingredient from one manufacturer shall be used in any one lot of starter slugs.

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 Starter slugs. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.2.2.3 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, level S-4, for duds, lagtime, intermittent smoke, flame, and deflagration. The sample for smoke emission time shall be equivalent to that segment of the sample specified in MIL-STD-414, level II (see 6.5).

4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample grenades and the levels A and B packaging shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.2.3.2 For starter slugs. The starter slugs shall be tested in accordance with 4.2.4.2 using an AQL of 0.40 percent.

4.2.3.3 For destructive tests. The entire sample shall be evaluated for the characteristics listed in (b), (c), (d) and (e) below and figure 1.

(a) For smoke emission time (see 6.5 and 6.6). Sample grenades shall be tested in accordance with 4.2.4 and MIL-STD-414, example B3 using an AQL of 2.5 percent defective.

(b) Duds. Sample grenades shall be tested in accordance with 4.2.4 and MIL-STD-105 using an AQL of 1.5 percent defective. Fuze failure shall be counted as a defect. The cause of the dud shall be identified (see 6.4).

(c) Lagtime, intermittent smoke and flame. Sample grenades shall be tested in accordance with 4.2.4 and MIL-STD-105 using an AQL of 2.5 percent defective.

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(d) **Deflagration.** Sample grenades shall be tested in accordance with 4.2.4. Any evidence of deflagration shall be cause for rejection of the lot.

(e) **Incorrect color.** Sample grenades shall be tested in accordance with 4.2.4. Any grenade which emits smoke of an entirely different color shall be cause for rejection of the entire lot.

4.2.3.3 For critical defects. Each grenade in the lot shall be examined for its critical characteristic listed in the classification of defects.

4.2.3.4 Classification of defects.

(a) Grenade, hand, smoke, HC, AN-M8 - assembly (Drawing 13-19-32)

Categories	Defects	Acceptance standards
Critical:		
1	Safety pin not spread	Visual
Major:	AQL 1.0 percent defective	
101	Filling height incorrect	4.2.4.3
Major:	AQL 1.0 percent defective	
102	Component missing or incorrectly assembled	
103	Incorrect torque applied to fuze	
104	Tape loose or does not completely cover hole	
105	Marking missing, illegible, or incorrect	
106	Workmanship (3.7)	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing or incorrect	

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(b) Starter cup assembly, prior to assembly (Drawing 13-19-94)

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Depth of cavity in slug incorrect
102	Diameter of cavity in slug incorrect
103	Height of slug incorrect
104	Slug weight incorrect
105	Slug cracked or chipped

(c) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Loose pack (including loose grenade)
102	Marking missing, incorrect, or illegible
103	Instruction card missing
104	Strap loose or missing

4.2.4 Tests.

4.2.4.1 Functioning. The grenade shall be submerged in water in an upright position for a minimum period of 30 minutes to a depth of 1 to 2 inches measured from the top surface of the grenade. The temperature of the water shall be colder or equal to but no greater than 10°F above the temperature of the grenade at the time of immersion. Within 1 hour after removal from the water, the grenade shall be functioned as follows: Grasp the grenade with the fuze lever against the palm; remove the safety pin and throw the grenade downwind in an area that is clear of flammable material, grass, etc. (see 6.6) that may influence flaming time observation if flaming occurs. The test results provided shall include the following as well as the standard test data:

(a) Lag time
(b) Total burning time

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(c) Total flaming time
(d) Smoke emission time

4.2.4.2 Starter slugs. The volatiles content of each lot of starter slugs shall be determined by drying the samples in a vacuum desiccator to constant weight.

4.2.4.3 Filling height. Examination for correct filling height shall be performed as an "in process" check on each lot using MIL-STD-105 with an AQL of 1.0 percent defective.

4.2.4.4 Moisture content. The moisture content of the pyromixture shall be measured prior to consolidation into grenades. When a batch fails to meet the moisture content requirement, it may be reblended or combined to bring the moisture content within drawing requirements.

5. PACKAGING

5.1 Unit packing, level A. Grenades shall be individually unit packed as specified on Drawing 13-9-120.

5.2 Packing. Level of packing shall be as specified (see 6.2).

5.2.1 Level A. Sixteen grenades unit packed as specified in 5.1 shall be packed together, fuze end up, in one layer, in an overseas-type or class wood shipping container conforming to either PPP-B-601, style J; PPP-B-621, class 2, style 4, type II; or type I, class 2, grade A of MIL-B-46506, except that one additional vertical batten shall be applied to the center of each end panel. The shipping container or the finished wood parts thereof shall be completely immersed for a minimum of one minute in a water-repellent wood preservative conforming to composition A of TT-W-572. Fiberboard pads, conforming to type CF, grade W5c, variety SW of PPP-F-320 shall completely separate the contents from the shipping container. Additional pads shall be used as required to insure a tight pack. Inside dimensions of the shipping container shall be approximately 12-1/2 by 12-1/2 by 6-5/8 inches with Drawing 13-9-120 (SH1 of 2 and SH2 of 2) Packaging Grenade Assembly. An instruction card conforming to Drawing 13-19-38 shall be placed on top of the packed grenades. The PPP-B-601 and PPP-B-621 shipping containers shall be closed and strapped in accordance with the applicable box specification using galvanized strapping. The MIL-B-46506 wirebound shipping container shall be closed in accordance with its specification and additionally secured with two galvanized flat straps, 0.015 inch by 3/8 inch applied girthwise around the top sides and bottom as close to the ends as practicable. Loop closures shall be firmly secured after application of the straps. Prior to packing the grenades in the PPP-B-601 and PPP-B-621 treated shipping containers, the preservative shall be dry as evidenced by the dryness test specified in MIL-B-2427. The MIL-B-46506 shipping containers shall be tested for dryness in accordance with the dryness test specified therein.

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5.2.2 Level B. Sixteen grenades unit packed as specified in 5.1 shall be packed as specified in 5.2.1 except that coated-finish strapping shall be used.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages shall be marked in accordance with Drawing 13-9-105 and Drawing 13-9-120. Shipping containers shall be marked in accordance with Drawing 13-19-83. Sufficient drying time shall be allowed after application of the wood preservative to insure that all marking is clear, legible, and permanent.

5.4 Palletization. Grenades packed as specified in 5.2.1 and 5.2.2 shall be palletized in accordance with in accordance with Drawing 19-48-4116/65.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This grenade is used for signaling purposes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)
(c) Level of packing required (see 5.2)
(d) Palletization when required

6.3 Directive. U.S. Army Chemical Research, Development and Engineering Center Directive No. 396 contains a suggested method of loading the grenade.

6.4 Dud. In the event of end item failure due to fuzes after a 15 minute waiting period, the grenades may be refuzed and retested to permit evaluation of the grenade for smoke emission time.

6.5 Definitions. (See figure 1 for representation of intervals.)

(a) Dud. A malfunction due to a mechanical or pyrotechnic failure.

(b) Lag time. The interval between the release of the fuze lever and the start of emission of smoke under pressure, including fuze delay time.

(c) Intermittent smoke. Cessation of smoke cloud under pressure for a period of 10 seconds or more followed by resumption of smoke cloud under pressure.

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(d) **Smoke emission time.** The period or periods of time, during the burning of the smoke mixture, when colored smoke is emitted under pressure.

(e) **Deflagration.** A sudden and violent increase in rate of burning which causes the grenade to rupture or separate, or which causes the violent expulsion of the solid contents or the metal parts of the grenade.

(f) **Burning time.** When the fuze lever is pulled off, the burning time is the period after the lag time [6.5(b)] until burning of the can contents under pressure ceases.

(g) **Flaming time.** The functioning of the grenade under pressure in a manner such that smoke emission time is interrupted by the presence of flame and/or gross reduction of desired HC smoke.

6.6 Caution. Precautions should be taken to prevent fires during preparation of the mixtures and while assembling grenades. Due to the hazard from exposure to chemicals, a dust respirator, coveralls, and gloves should be worn when handling these materials. Caution should also be exercised during the functioning test as a hazard exists from flying hot particles and possible detonation resulting from incorrect mixture.

6.7 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.8 Component specifications. The following specifications reflect the components required for the completion of the AN-M8 hand grenade:

MIL-G-1353 - Grenades, Hand; Inert Components for
MIL-F-51353 - Fuze, Hand Grenade, M201A1

6.9 Subject term (key word) listing.

Grenade, Fuze
Starter Cup Assembly
Starter Slug
White Smoke

6.10 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous isssue due to the extensiveness of the changes.

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Navy - MC
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-12327J	GRENADE, HAND, SMOKE, HC, M8

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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